boundary conditions: 1. An insulated boundary along the edge AD, 2. A given temperature of 100^oC at the edge AB, 3. A surface convection to ambient temperature of 0^oC along the edge BC and DC. It is noted that the severe discontinuity in boundary conditions at point B gives rise to a quasi-singular type behaviour at this location. This fact, combined with the relatively coarse meshes specified in [15], made it almost impossible to obtain a good comparison to the target value of temperature at point E which lies at a distance of $0.2m$ above B on the edge BC. The analytical solution for the temperature at this point is 18.2535^oC.

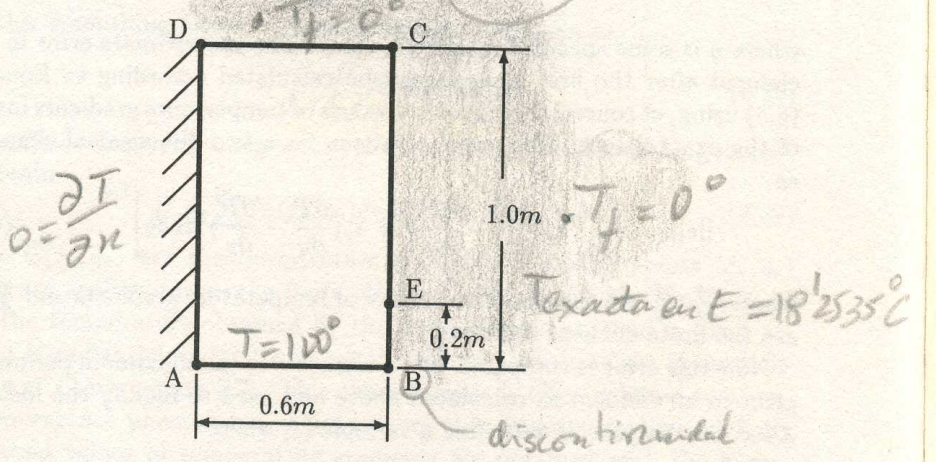


Figure 6.1: Two dimensional heat transfer with convection

The analytical solution obtained from solving the differential equation with the specified boundary conditions is derived in [16] and is given by

$$T = 2hT_{AB}\sum_{n=1}^{\infty}\frac{cos\alpha_n x[\alpha_n cosh\alpha_n(b-y) + \underline{h}sinh\alpha_n(b-y)]}{[(\alpha_n^2+\underline{h}^2)a+\underline{h}](\alpha_n cosh\alpha_n b + \underline{h}sinh\alpha_n b)cos\alpha_n a} \quad (6.34)$$

where

$$\begin{aligned} a &= 0.6m \\ b &= 1.0m \\ \underline{h} &= h/k \\ k &= \text{thermal conductivity} = 52\ W/m^oC \\ h &= \text{heat transfer coefficient} = 750\ W/m^oC \\ T_{AB} &= 100^oC \end{aligned}$$

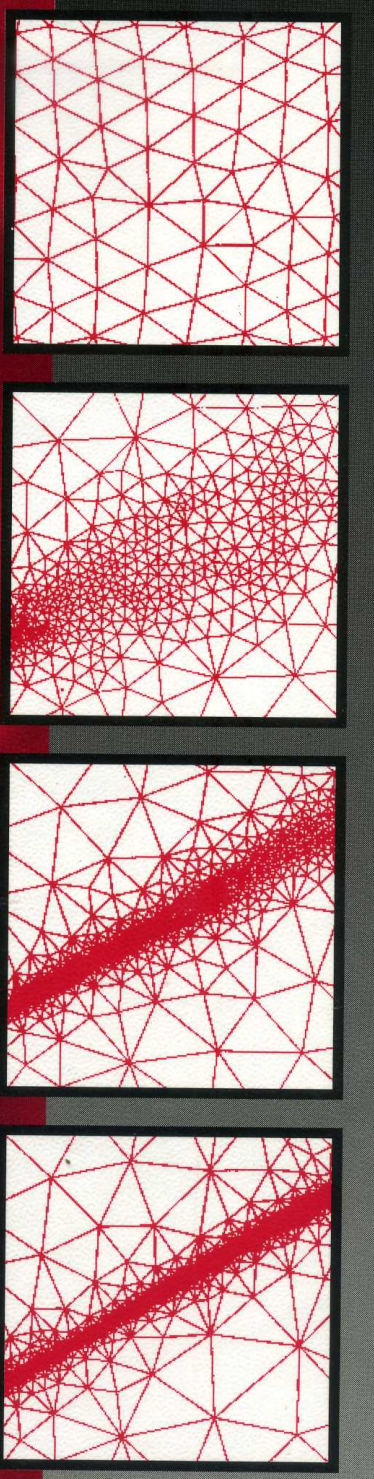
and α_n are the roots of "$\alpha\ tan\ \alpha a = \underline{h}$" which may be solved using the iterative method as follows.

$$\alpha_n = \frac{1}{a}[\tan^{-1}(\underline{h}/\alpha_n) + (n-1)\pi]$$

Hou-Cheng Huang
Asif S. Usmani

Finite Element Analysis for Heat Transfer

Theory and Software



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Cover illustration: Chapter 7, Figure 5. Sequence of automatically-generated meshes based on a specified target error of 20%.

ISBN 3-540-19880-6 Springer-Verlag Berlin Heidelberg New York
ISBN 0-387-19880-6 Springer-Verlag New York Berlin Heidelberg

British Library Cataloguing in Publication Data
A catalogue record for this book is available from the British Library

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Printed in Great Britain

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Typesetting: Camera ready produced by authors using LaTeX
Printed and bound at the Athenæum Press Ltd, Gateshead, Tyne & Wear.
69/3830-543210 Printed on acid-free paper